PHYS101/Fall 2013-2014 Final Exam

Physics 101 No: _____

Name: _____

(On my honor I have neither given nor received any unauthorized assistance on this exam.)

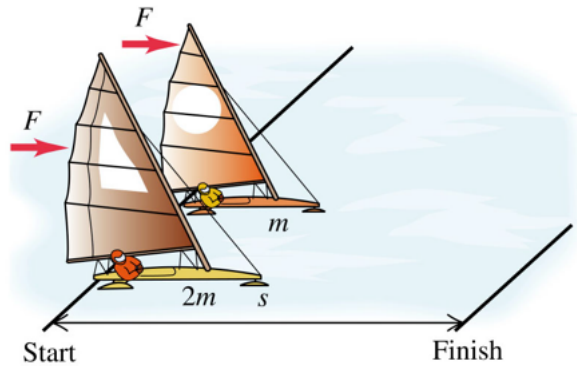
- This exam is for 120 minutes and has two questions and four problems (Check this now carefully.)
- This is a closed-book/notes/calculator/laptop exam.
- Read the questions carefully and write down your solutions in the space provided neatly and clearly. Write down your final results in the boxes at the end of each question.
- Answers without work shown will not be given any credit.
- A formula sheet is provided for you.

(Leave blank)

Question 1	Question 2	Problem 1	Problem 2	Problem 3	Problem 4	Total

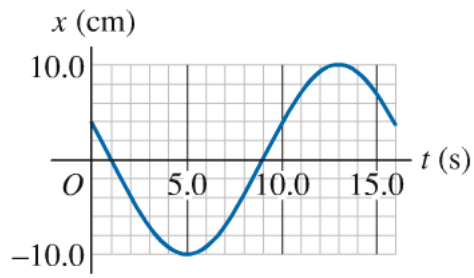
Question 1 (10 points)

Two iceboats (one of mass m, one of mass 2m) hold a race on a frictionless, horizontal, frozen lake. Both iceboats start at rest, and the wind exerts the same constant force on both iceboats. (a) Which iceboat crosses the finish line with more kinetic energy? (b) Which iceboat crosses the finish line with more momentum?



Question 2 (10 points)

A small block of mass $m = 1$ kg is attached to an ideal spring. The displacement of the system as a function of time is shown in the figure. What are



(a) (2 pts.) the amplitude?

(b) (2 pts.) the period?

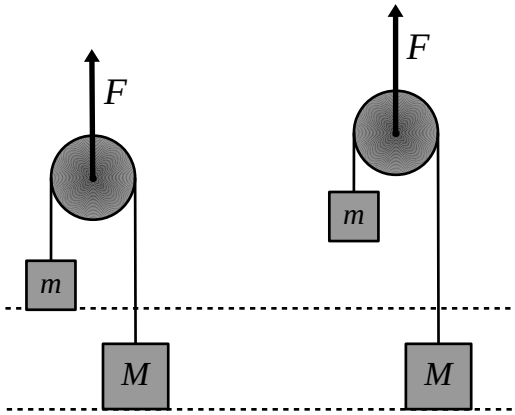
(c) (3 pts.) the angular frequency?

(c) (3 pts.) the spring constant?

Problem 1 (25 points)

Two masses, m =3 kg and M = 6 kg are connected with a light rope passing over a massless pulley. A vertically upward force with magnitude **F** is applied to pulley's axle as shown in the figure. It is observed that the 6 kg mass doesn't change its position.

(You may take $g = 10.0$ m/s^2)

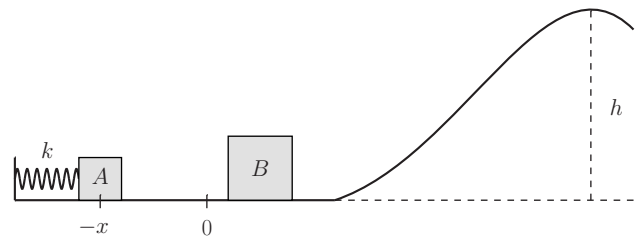


(a) (13 pts.) Find the tension in the rope.

(b) (12 pts.) Find the acceleration of the 3 kg mass.

Problem 2 (25 points)

Block A with mass 3.6 kg is attached to one end of a massless spring with force constant 810 N/m which is fixed on a wall from the other end. The spring is compressed a distance of x and block B with mass 7.2 kg is placed on a position such that when the spring is released block A collides with block B at the equilibrium position of the spring. The collision is **elastic**, and block B climbs up on a hill with maximum altitude of 180 cm after the collision. (take $g = 10.0\text{m/s}^2$)

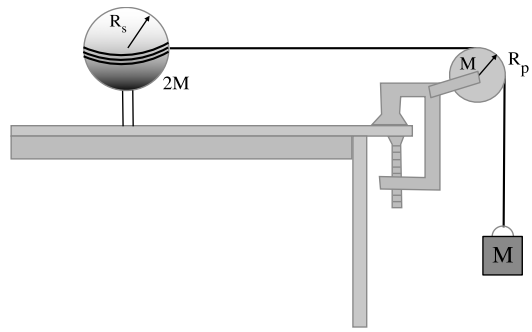


(a) (15 pts.) What must be the minimum value of compression (x) so that block B can pass over the hill's highest point?

(b) (10 pts.) Setting $t = 0$ at the instant of collision, write down the position function of block A after the collision and find the speed of it at $t = 4.0$ s.

Problem 3 (25 points)

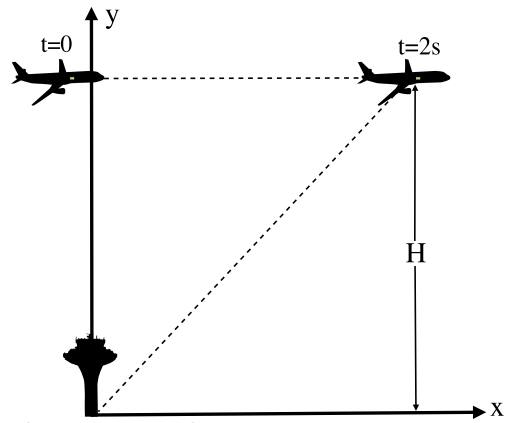
A uniform spherical shell of radius R_s can rotate about a vertical axis on the frictionless bearings. A massless rope passes around the equator of the shell, over a pulley of radius R_p and is attached to a small block. There is no friction on the pulley's axle and the rope does not slip on the pulley. The pulley and the small block have the same mass M and the sphere has mass $2M$. Find the speed of the block after the block has fallen 11.5 cm.



($I_{\text{sphere}} = \frac{2}{5}mr^2$, $I_{\text{disk}} = \frac{1}{2}mr^2$, $g = 10.0$ m/s^2)

Problem 4 (25 points)

A 5000 kg airplane is flying in straight horizontal line 50 m above the ground with a velocity given by $\vec{v}(t) = b\,t^3\ \hat{\imath}$, where $b = 4\text{m/s}^4$ with t in seconds. At $t = 0$ the plane is just above the airport tower. At $t = 2$ s, <u>in unit vector notation</u>



(a) (5 pts.) Find the position vector of the plane taking the tower at the origin.

$\vec{r}=$

(b) (5 pts.) Find the instantaneous acceleration of the plane.

$\vec{a}=$

(c) (5 pts.) Find the angular momentum of the plane about the tower.

$\vec{L}=$

(d) (5 pts.) Find the torque on the plane about the tower.

$\vec{\tau}=$

Formula Sheet

Scalar Product:

$c = \vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$

$c = AB \cos\phi$

Vector Product:

$\vec{C} = \vec{A} \times \vec{B}$

$C = AB \sin\phi$

$C_x = A_y B_z - A_z B_y$

$C_y = A_z B_x - A_x B_z$

$C_z = A_x B_y - A_y B_x$

Straight-line motion:

$$v_{av-x} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\boldsymbol{\Delta x}}{\boldsymbol{\Delta t}}$$

$$v_x = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta x}}{\boldsymbol{\Delta t}} = \frac{dx}{dt}$$

$$a_{av-x} = \frac{v_{2x} - v_{1x}}{t_2 - t_1} = \frac{\boldsymbol{\Delta v_x}}{\boldsymbol{\Delta t}}$$

$$a_x = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta v_x}}{\boldsymbol{\Delta t}} = \frac{dv_x}{dt}$$

Constant x-acceleration:

$v_x = v_{0x} + a_x t$

$x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2$

$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$

$x - x_0 = \left(\frac{v_{0x} + v_x}{2}\right) t$

Motion in three dimensions:

$\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$

$$\vec{v}_{av} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} = \frac{\boldsymbol{\Delta \vec{r}}}{\boldsymbol{\Delta t}}$$

$$\vec{v} = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta \vec{r}}}{\boldsymbol{\Delta t}} = \frac{d\vec{r}}{dt}$$

$$\vec{a}_{av} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} = \frac{\boldsymbol{\Delta \vec{v}}}{\boldsymbol{\Delta t}}$$

$$\vec{a} = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta \vec{v}}}{\boldsymbol{\Delta t}} = \frac{d\vec{v}}{dt}$$

Circular motion:

$$a_{\text{rad}} = \frac{v^2}{R} \quad a_{\text{rad}} = \frac{4\pi^2 R}{T^2}$$

Relative velocity:

$v_{\text{P/A-}x} = v_{\text{P/B-}x} + v_{\text{B/A-}x}$

$\vec{v}_{\text{P/A}} = \vec{v}_{\text{P/B}} + \vec{v}_{\text{B/A}}$

Newton's 1st Law:

$\vec{R} = \vec{F}_1 + \vec{F}_2 + \ldots = \sum \vec{F}$

$\sum \vec{F} = 0$

Newton's 2nd Law:

$\sum \vec{F} = m\vec{a}$

Newton's 3rd Law:

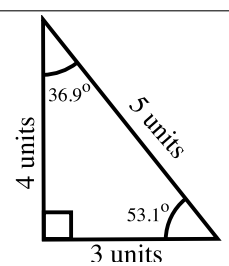
$\vec{F}_{\text{A on B}} = -\vec{F}_{\text{B on A}}$

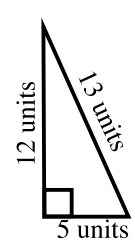
Friction Force:

$f_k = \mu_k n \quad f_s \leq \mu_s n$

$\sin 36.9^\circ = \cos 53.1^\circ = 0.6$

$\sin 53.1^\circ = \cos 36.9^\circ = 0.8$





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Signature:

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Friction Force:

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Work done by a force:

$W = \vec{F} \cdot \vec{s} = Fs\cos\phi$

Kinetic energy:

$K = \dfrac{1}{2} m v^2$

The work-energy theorem:

$W_{\text{tot}} = K_2 - K1 = \Delta K$

Work done by a varying force:

$W = \int_{x_1}^{x_2} F_x \; dx$

Power:

$P_{\text{av}} = \dfrac{\Delta W}{\Delta t}$

$P_{\text{av}} = \vec{F} \cdot \vec{v}$

Gravitational and elastic pot. energy:

$W_{\text{grav}} = mgy_1 - mgy_2$

Momentum of a particle:

$\vec{p} = m\vec{v}$

$\sum \vec{F} = \dfrac{d\vec{p}}{dt}$

Impulse:

$\vec{J} = \sum \vec{F} \Delta t$

$\vec{J} = \vec{p}_2 - \vec{p}_1$

$W_{\text{el}} = \dfrac{1}{2} k x_1^2 - \dfrac{1}{2} k x_2^2$

Conservation of mech. energy:

$K_1 + U_1 + W_{\text{other}} = K_2 + U_2$

Determining force from potential:

$\vec{F} = -\left(\dfrac{\partial U}{\partial x}\hat{\imath} + \dfrac{\partial U}{\partial y}\hat{\jmath} + \dfrac{\partial U}{\partial z}\hat{k}\right)$

Conservation of momentum:

$\vec{P} = \vec{p}_A + \vec{p}_B + \ldots$

Elastic collision:

$v_{B2x} = \dfrac{2m_A}{m_A + m_B} v_{A1x}$ (assuming $v_{B1x} = 0$)

Center of mass:

$\vec{r}_{\text{cm}} = \dfrac{m_1\vec{r}_1 + m_2\vec{r}_2 + \ldots}{m_1 + m_2 + \ldots}$

Rotational kinematics:

$\omega_z = \dfrac{d\theta}{dt}$

$\alpha_z = \dfrac{d\omega_z}{dt}$

$\theta = \theta_0 + \omega_0 t + \dfrac{1}{2}\alpha_z t^2$

$v = r\omega$

$a_{\tan} = r\alpha$

$a_{\text{rad}} = \dfrac{v^2}{r} = \omega^2 r$

Torque:

$\vec{\tau} = \vec{r} \times \vec{F}$

Rotational dynamics:

$\sum \tau_z = I\alpha_z$

$K = \dfrac{1}{2} M v_{\text{cm}}^2 + \dfrac{1}{2} I_{\text{cm}} \omega^2$

Rolling without slipping:

$v_{\text{cm}} = R\omega$

Periodic motion:

$f = \dfrac{1}{T}, \qquad \omega = 2\pi f$

Simple harmonic motion:

Spring: $F_x = -kx, \qquad \omega = \sqrt{\dfrac{k}{m}}$

$x = A \;\cos(\omega t + \phi)$

Simple pendulum: $\omega = \sqrt{\dfrac{g}{L}}$

Damped oscillations:

$x = Ae^{(-b/2m)t} \;\cos(\omega' t + \phi)$

$\omega' = \sqrt{\dfrac{k}{m} - \dfrac{b^2}{4m^2}}$

Mechanical waves:

$v = \lambda f = \omega / k$

$y(x,t) = A \;\cos(kx - \omega t)$

$k = 2\pi/\lambda, \;\; \omega = 2\pi f = v$

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